

Aim: To study SQL commands, their uses, syntax, and install MySQL for database management experiments.

1. What is SQL and its Use?

- **SQL (Structured Query Language)** is a standard programming language used to create, manage, and manipulate databases. It allows users to store, retrieve, update, and delete data in a database.

Uses of SQL:

- Creating and managing databases and tables.
- Inserting new data into tables.
- Retrieving required data using queries.
- Updating existing data.
- Deleting unwanted data.
- Controlling access and permissions to databases.
- Performing operations such as sorting, filtering, and joining data from multiple tables.

2. Describe all the different commands of SQL, mention use of it and syntax.

- SQL commands are generally divided into **five main categories**: DDL, DML, DQL, DCL, and TCL.

1. DDL – Data Definition Language

DDL commands are used to define and modify the structure of a database and its tables.

Command	Use	Syntax
CREATE	Creates a database or table.	CREATE TABLE table_name (column1 datatype, column2 datatype);
ALTER	Modifies the structure of an existing table.	ALTER TABLE table_name ADD column_name datatype;

DROP	Permanently deletes a table or database.	DROP TABLE table_name;
TRUNCATE	Deletes all records from a table while keeping its structure.	TRUNCATE TABLE table_name;
RENAME	Changes the name of a table.	RENAME TABLE old_name TO new_name;

Example:

```
CREATE TABLE Student (
    RollNo INT,
    Name VARCHAR(50),
    Marks INT
);
```

2. DML – Data Manipulation Language

DML commands are used to insert, modify, and delete data stored in tables.

Command	Use	Syntax
INSERT	Adds new records to a table.	INSERT INTO table_name VALUES (value1, value2);
UPDATE	Modifies existing records.	UPDATE table_name SET column_name = value WHERE condition;
DELETE	Removes records from a table.	DELETE FROM table_name WHERE condition;

Example:

```
INSERT INTO Student VALUES (1, 'Rahul', 85);
```

```
UPDATE Student SET Marks = 90 WHERE RollNo = 1;
```

```
DELETE FROM Student WHERE RollNo = 1
```

3. DQL – Data Query Language

DQL is used to retrieve data from a database.

Command	Use	Syntax
SELECT	Retrieves data from one or more tables.	SELECT column1, column2 FROM table_name WHERE condition;

Example:

SELECT * FROM Student;

To retrieve students who scored more than 80:

SELECT Name, Marks

FROM Student

WHERE Marks > 80;

4. DCL – Data Control Language

DCL commands are used to **control access and permissions** in a database.

Command	Use	Syntax
GRANT	Gives privileges to a user.	GRANT privilege ON database_name.table_name TO user_name;
REVOKE	Removes privileges from a user.	REVOKE privilege ON database_name.table_name FROM user_name;

Example:

GRANT SELECT ON college.Student TO 'user1';

REVOKE SELECT ON college.Student FROM 'user1';

5. TCL – Transaction Control Language

TCL commands are used to **manage transactions in a database**.

Example:

UPDATE Student SET Marks = 95 WHERE RollNo = 1;

SAVEPOINT sp1;

UPDATE Student SET Marks = 100 WHERE RollNo = 1;

ROLLBACK TO SAVEPOINT sp1;

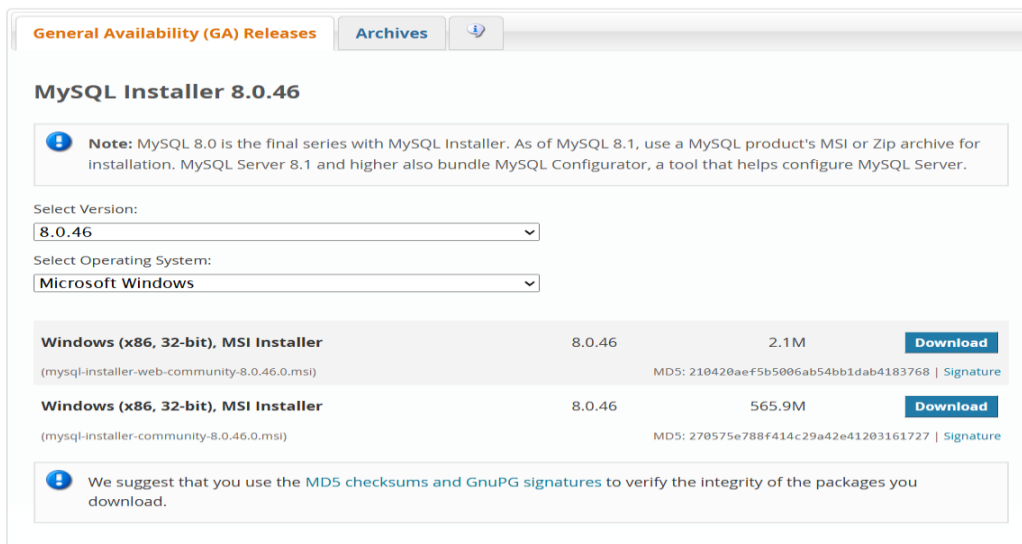
COMMIT;

Summary

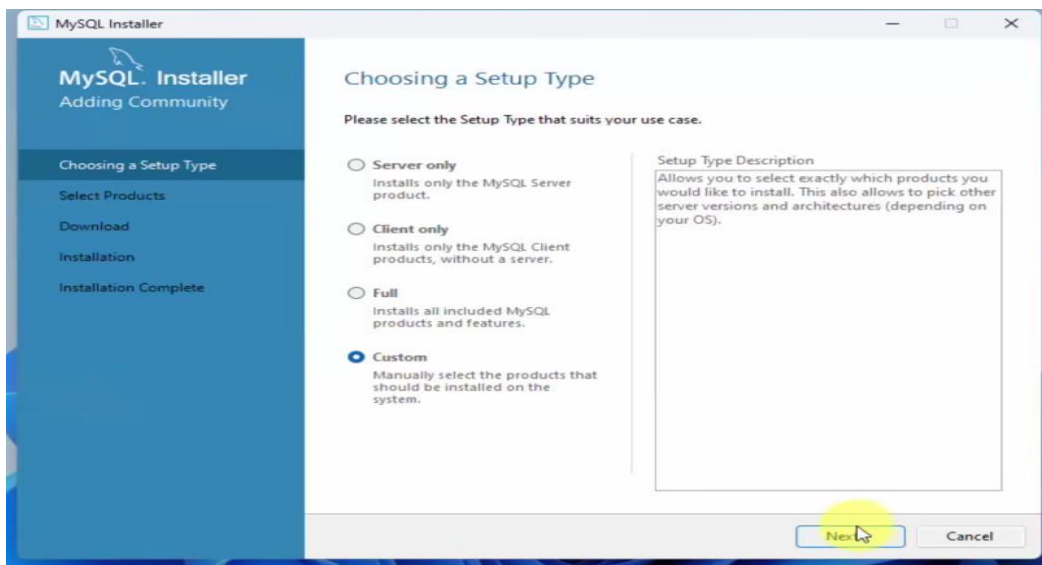
Category	Full Form	Main Commands	Purpose
DDL	Data Definition Language	CREATE, ALTER, DROP, TRUNCATE, RENAME	Defines database structure
DML	Data Manipulation Language	INSERT, UPDATE, DELETE	Manipulates data
DQL	Data Query Language	SELECT	Retrieves data
DCL	Data Control Language	GRANT, REVOKE	Controls user access
TCL	Transaction Control Language	COMMIT, ROLLBACK, SAVEPOINT	Manages transactions

3. Write a step by step procedure to install MYSQL.

1. Go to Web Browser and search for 'MySQL' and go to the web page of MySQL ,
“MYSQL community downloads” —> “MYSQL installer for windows

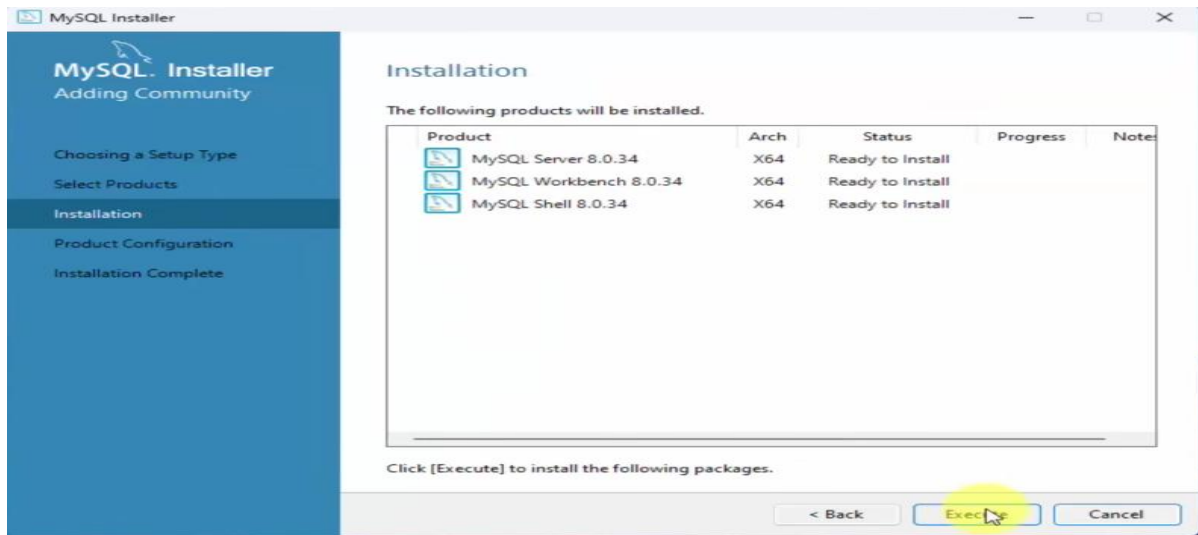


2. Run the Installer, for the setup type, select Custom. And “NEXT”.

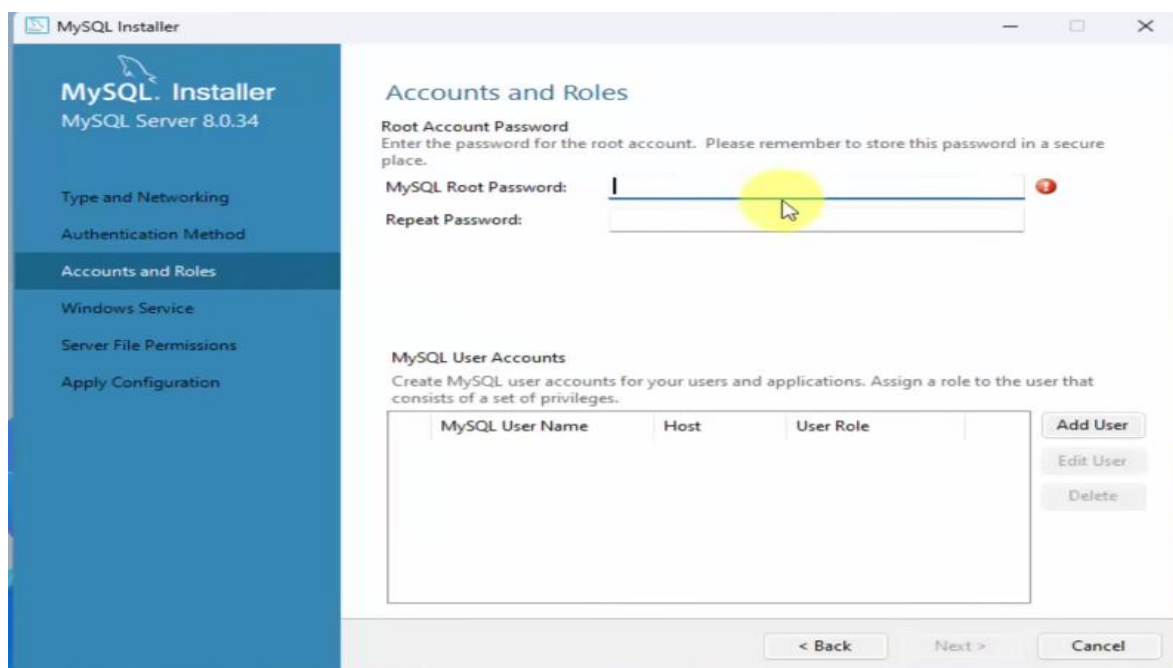


3. Select the Required Products

- Add the following products to the Products to be Installed section:
 - i) MySQL Server
 - ii) MySQL Workbench
 - iii) MySQL Shell

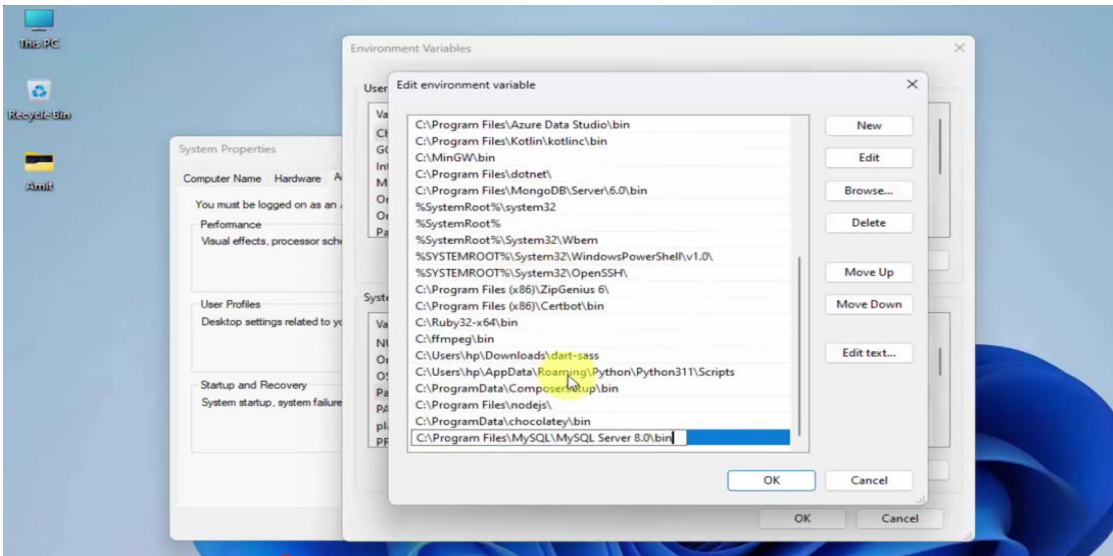


4. Configure Network Settings , Keep the default setting and keep clicking on next Then for authentication write a strong password, then click on next.



5. Keep clicking on NEXT , and then FINISH. Then , Set Environment Variables

- Copy the path of the MySQL bin folder (e.g., C:\Program Files\MySQL\MySQL Server\bin).
- Search for Environment Variables in the Windows Start menu.
- Under System Variables, select Path and click Edit.
- Click New, paste the copied path, and click OK.



6. Verify the Installation

- Open Command Prompt (CMD).
- Type the following command: `mysql --version`

```
Command Prompt
Microsoft Windows [Version 10.0.22621.1992]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hp>mysql --version
mysql Ver 8.0.34 for Win64 on x86_64 (MySQL Community Server - GPL)

C:\Users\hp>|
```

